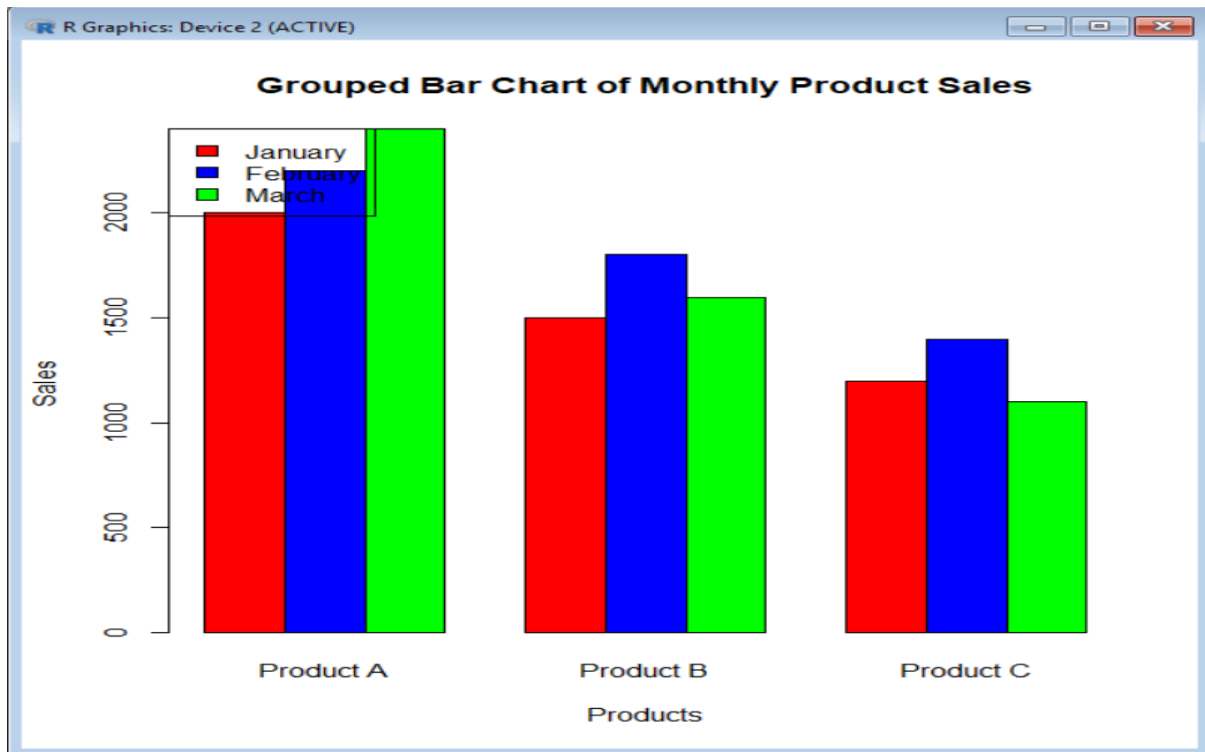


Set 6

Program 1-Grouped Bar Chart

```
product <- data.frame(  
  Product = c("Product A", "Product B", "Product C"),  
  January = c(2000, 1500, 1200),  
  February = c(2200, 1800, 1400),  
  March = c(2400, 1600, 1100)  
)  
sales <- t(as.matrix(product[,2:4]))  
barplot(  
  sales,  
  beside = TRUE,  
  col = c("red", "blue", "green"),  
  names.arg = product$Product,  
  main = "Grouped Bar Chart of Monthly Product Sales",  
  xlab = "Products",  
  ylab = "Sales"  
)  
legend(  
  "topleft",  
  legend = c("January", "February", "March"),  
  fill = c("red", "blue", "green")  
)
```

Output:

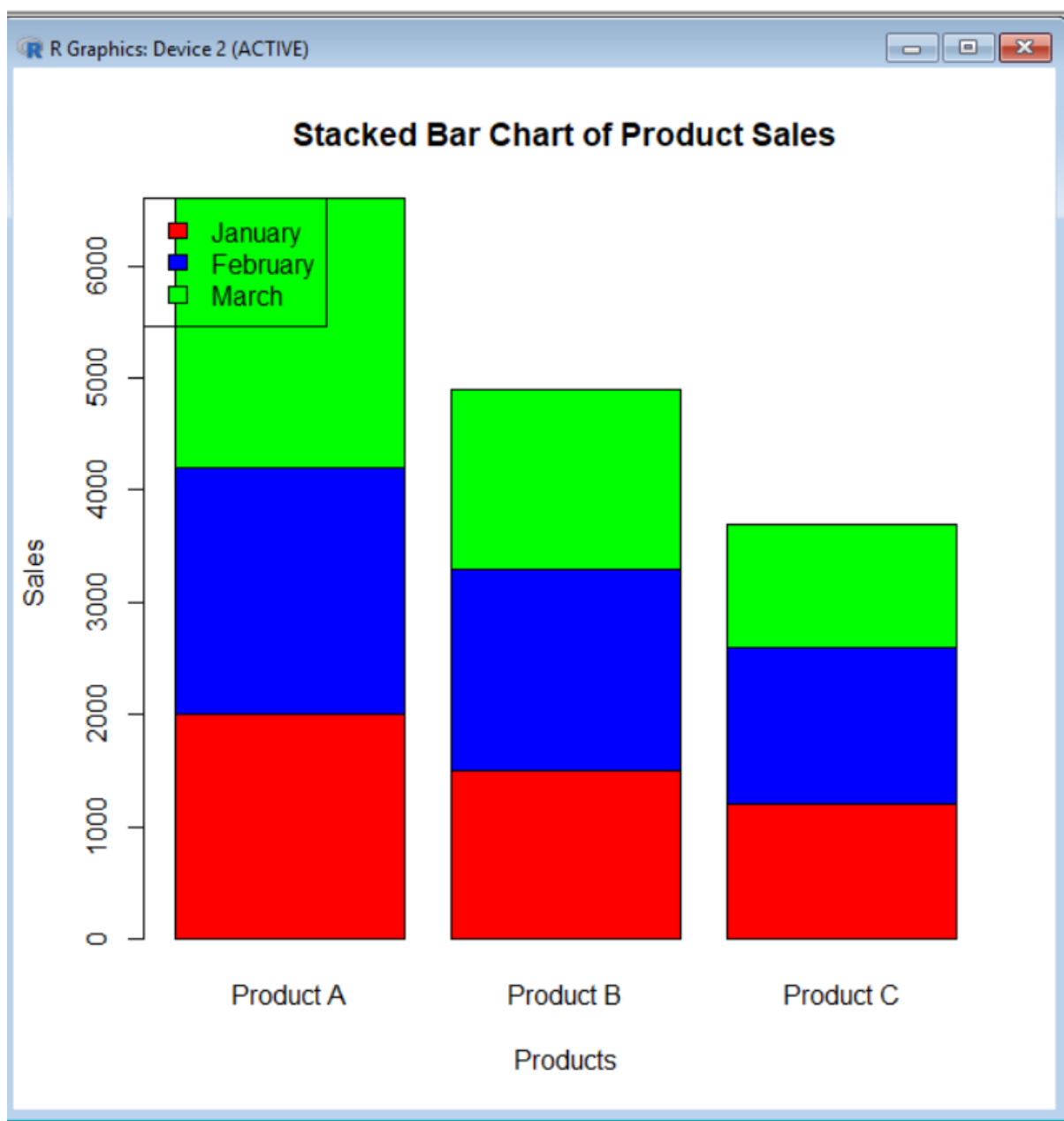


Program 2-Stacked Bar Chart

```
product <- data.frame(  
  Product = c("Product A", "Product B", "Product C"),  
  January = c(2000, 1500, 1200),  
  February = c(2200, 1800, 1400),  
  March = c(2400, 1600, 1100)  
)  
sales <- t(as.matrix(product[,2:4]))  
barplot(  
  sales,  
  beside = FALSE,  
  col = c("red", "blue", "green"),  
  names.arg = product$Product,  
  main = "Stacked Bar Chart of Product Sales",  
  xlab = "Products",  
  ylab = "Sales"
```

```
)  
legend(  
  "topleft",  
  legend = c("January", "February", "March"),  
  fill = c("red", "blue", "green")  
)
```

Output:



Program 3-Display Data Table

```
product <- data.frame(  
  Product_ID = c(1,2,3),  
  Product_Name = c("Product A","Product B","Product C"),  
  January = c(2000,1500,1200),  
  February = c(2200,1800,1400),  
  March = c(2400,1600,1100)  
)  
print("Monthly Product Sales Table")  
print(product)
```

Output:

	Product_ID	Product_Name	January	February	March
1	1	Product A	2000	2200	2400
2	2	Product B	1500	1800	1600
3	3	Product C	1200	1400	1100

Program 4-Total Sales Calculation

```
product <- data.frame(  
  Product_Name = c("Product A","Product B","Product C"),  
  January = c(2000,1500,1200),  
  February = c(2200,1800,1400),  
  March = c(2400,1600,1100)  
)  
product$Total_Sales <- rowSums(product[,2:4])  
print(product)  
cat("\nHighest Selling Product:",  
  product$Product_Name[which.max(product$Total_Sales)])
```

Output:

```
> cat("\nHighest Selling Product:",  
+     product$Product_Name[which.max(product$Total_Sales)])
```

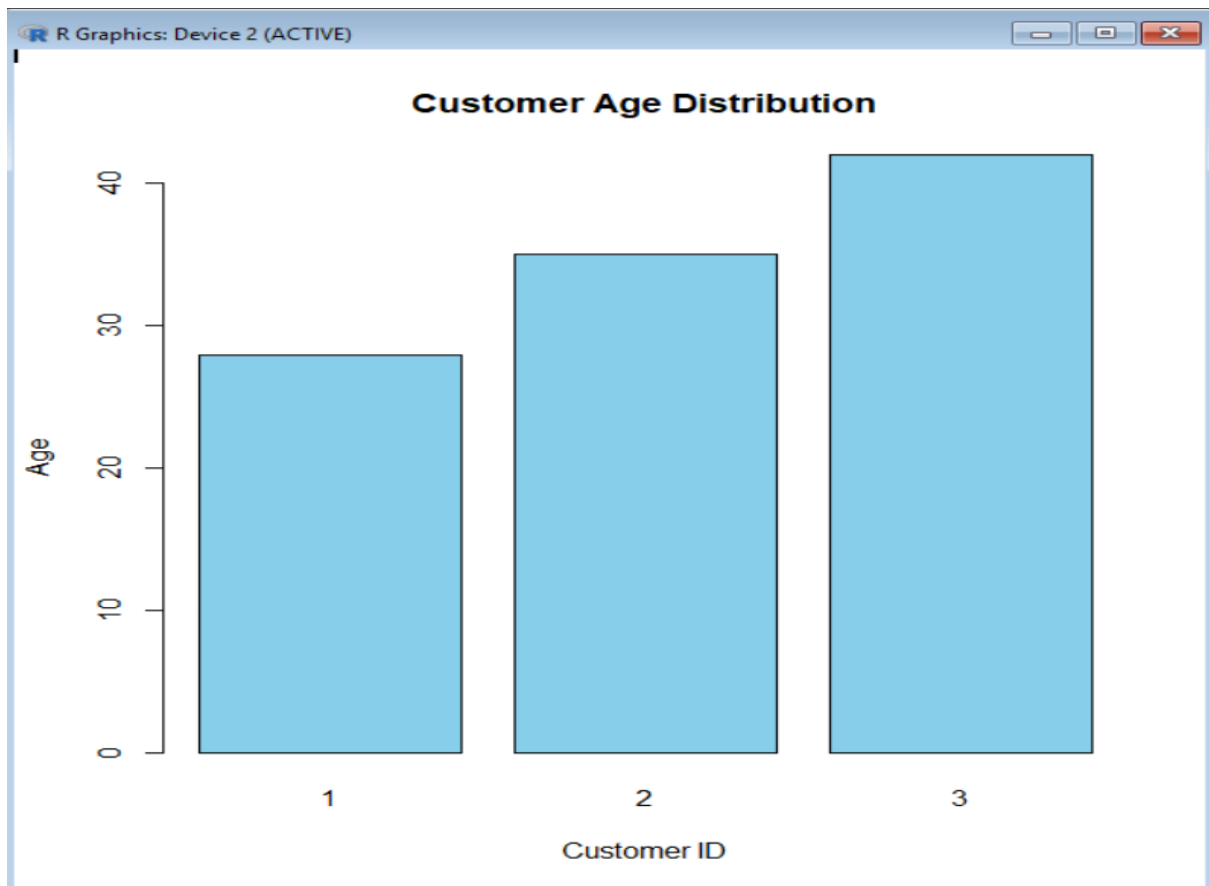
```
Highest Selling Product: Product A> |
```

Set 7

Program 1 - Bar Chart (Customer Ages)

```
customer <- data.frame(  
  Customer_ID = c(1,2,3),  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)  
barplot(  
  customer$Age,  
  names.arg = customer$Customer_ID,  
  col = "skyblue",  
  main = "Customer Age Distribution",  
  xlab = "Customer ID",  
  ylab = "Age"  
)
```

Output:

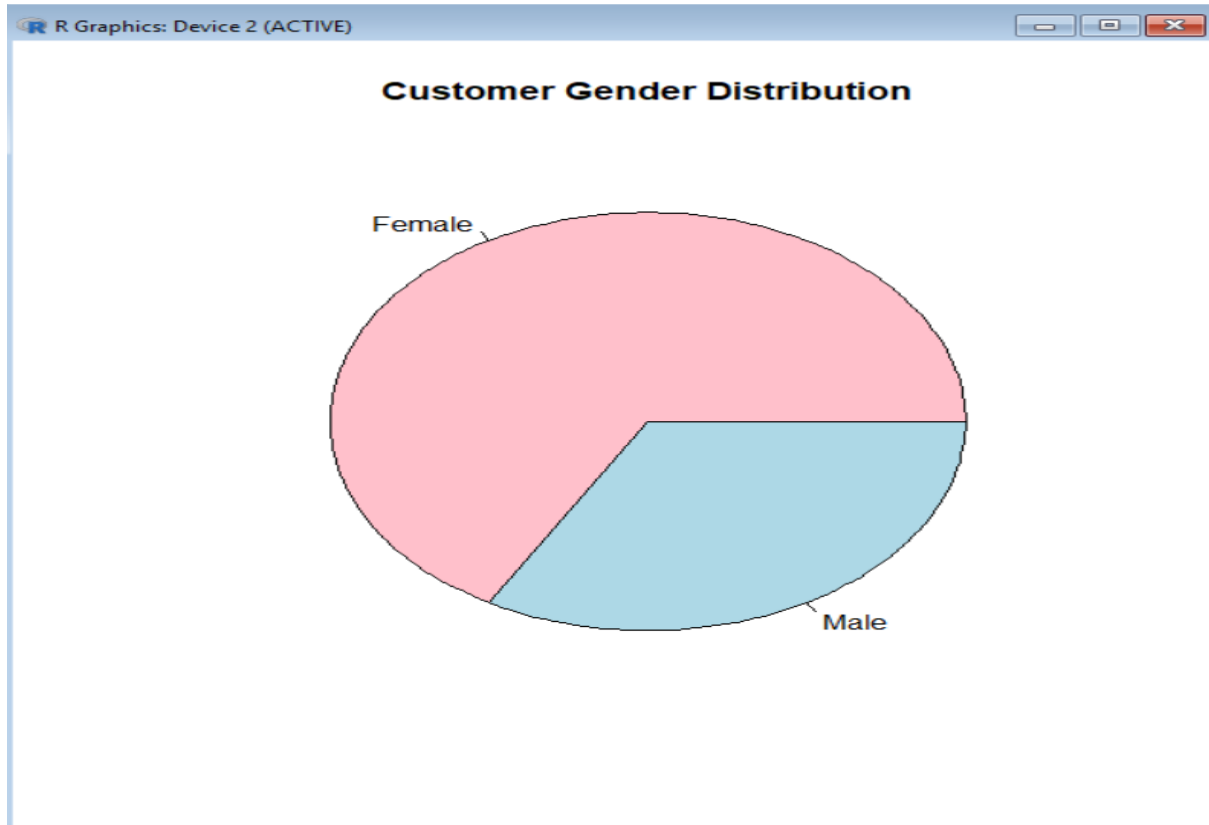


Program 2 - Pie Chart (Gender Distribution)

```
customer <- data.frame(  
  Customer_ID = c(1,2,3),  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)  
gender_count <- table(customer$Gender)  
pie(  
  gender_count,  
  col = c("pink","lightblue"),  
  main = "Customer Gender Distribution"
```

)

Output:



Program 3 - Display Data Table

```
customer <- data.frame(  
  Customer_ID = c(1,2,3),  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)
```

```
print("Customer Demographics Table")
```

```
print(customer)
```

Output:

```
> print(customer)
  Customer_ID Age Gender Income
1           1  28 Female  50000
2           2  35   Male  60000
3           3  42 Female  75000
```

Program 4 – Summary of Customer Data

```
customer <- data.frame(
  Customer_ID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
cat("Average Age:", mean(customer$Age), "\n")
cat("Average Income:", mean(customer$Income), "\n")
print(table(customer$Gender))
```

Output:

```
> print(table(customer$Gender))
```

```
Female   Male
      2      1
```


Set 8

Program 1 - Line Chart (Performance Trend)

```
employee <- data.frame(  
  Employee_ID = c(1,2,3),  
  Department = c("Sales","HR","Marketing"),  
  Years_Service = c(5,3,7),  
  Performance_Score = c(85,92,78)  
)  
  
plot(  
  employee$Years_Service,  
  employee$Performance_Score,  
  type = "o",  
  col = "blue",  
  pch = 16,  
  main = "Employee Performance Trend",  
  xlab = "Years of Service",  
  ylab = "Performance Score"  
)
```

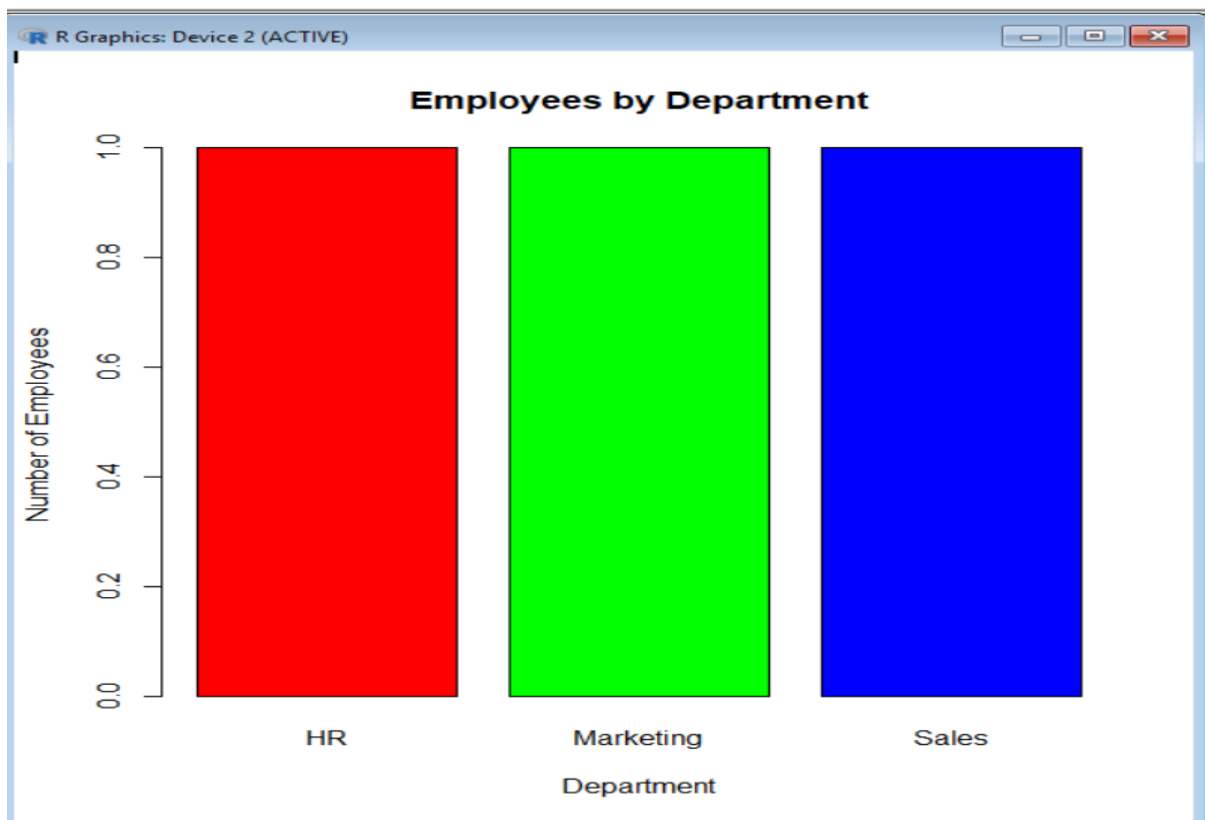
Output:



Program 2 - Bar Chart (Employees Across Departments)

```
employee <- data.frame(  
  Employee_ID = c(1,2,3),  
  Department = c("Sales","HR","Marketing"),  
  Years_Service = c(5,3,7),  
  Performance_Score = c(85,92,78)  
)  
dept_count <- table(employee$Department)  
barplot(  
  dept_count,  
  col = c("red","green","blue"),  
  main = "Employees by Department",  
  xlab = "Department",  
  ylab = "Number of Employees"  
)
```

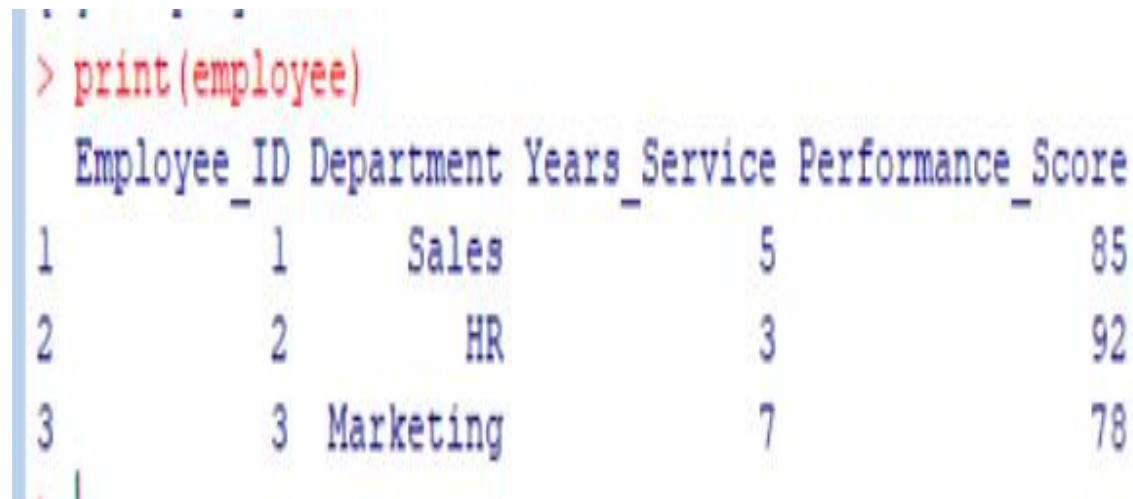
Output:



Program 3 - Display Data Table

```
employee <- data.frame(  
  Employee_ID = c(1,2,3),  
  Department = c("Sales","HR","Marketing"),  
  Years_Service = c(5,3,7),  
  Performance_Score = c(85,92,78)  
)  
print("Employee Performance Table")  
print(employee)
```

Output:



```
> print(employee)  
Employee_ID Department Years_Service Performance_Score  
1          1      Sales             5             85  
2          2        HR             3             92  
3          3  Marketing             7             78
```

Program 4 -Employee Performance Summary

```
employee <- data.frame(  
  Employee_ID = c(1,2,3),  
  Department = c("Sales","HR","Marketing"),  
  Years_Service = c(5,3,7),  
  Performance_Score = c(85,92,78)  
)  
cat("Average Performance Score:",  
    mean(employee$Performance_Score), "\n")
```

```
cat("Average Years of Service:",
    mean(employee$Years_Service), "\n")
cat("Best Performing Employee ID:",
    employee$Employee_ID[which.max(employee$Performance_Score)], "\n")
```

Output:

```
> # Employee Performance Dataset
> employee <- data.frame(
+   Employee_ID = c(1,2,3),
+   Department = c("Sales","HR","Marketing"),
+   Years_Service = c(5,3,7),
+   Performance_Score = c(85,92,78)
+ )
>
> cat("Average Performance Score:",
+     mean(employee$Performance_Score), "\n")
Average Performance Score: 85
>
> cat("Average Years of Service:",
+     mean(employee$Years_Service), "\n")
Average Years of Service: 5
>
> cat("Best Performing Employee ID:",
+     employee$Employee_ID[which.max(employee$Performance_Score)], "\n")
Best Performing Employee ID: 2
```

Srt 9

Program 1-Histogram and Box Plot

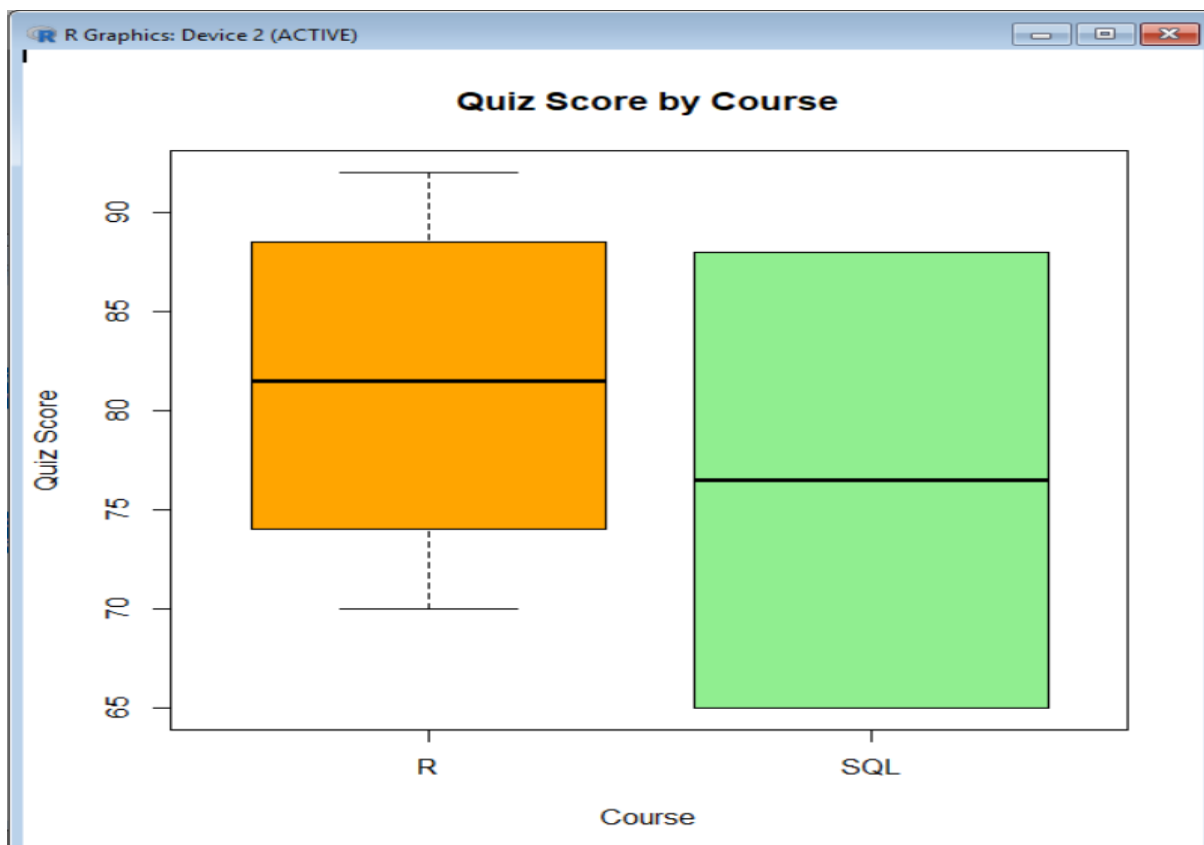
```
data <- data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Course = c("R","R","SQL","R","R","SQL"),
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
  Videos_Watched = c(12,15,8,18,9,14),
```

```

Quiz_Score = c(78,85,65,92,70,88),
Login_Date = c("2025-01-05","2025-01-05","2025-02-08",
               "2025-02-08","2025-03-12","2025-03-12")
)
hist(data$Quiz_Score,
     col="skyblue",
     main="Histogram of Quiz Scores",
     xlab="Quiz Score")
boxplot(Quiz_Score ~ Course,
       data=data,
       col=c("orange","lightgreen"),
       main="Quiz Score by Course",
       xlab="Course",
       ylab="Quiz Score")

```

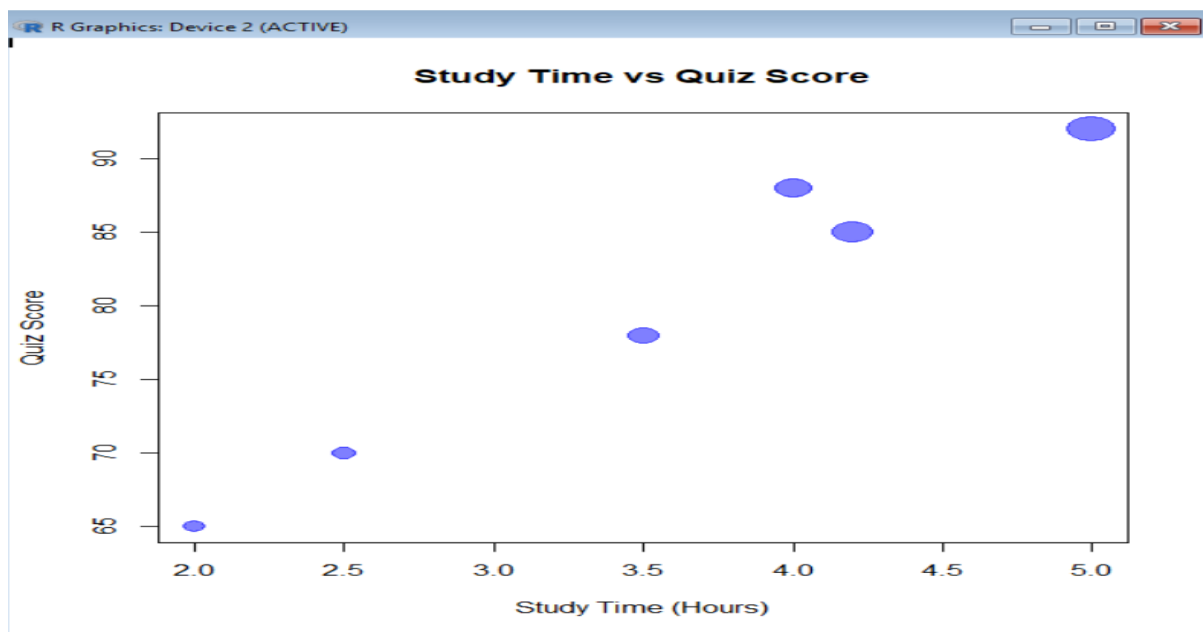
Output:



Program 2-Scatter Plot with Bubble Size

```
data <- data.frame(  
  Student_ID = c("L01","L02","L03","L04","L05","L06"),  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Age = c(20,22,19,21,23,20),  
  Course = c("R","R","SQL","R","R","SQL"),  
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),  
  Videos_Watched = c(12,15,8,18,9,14),  
  Quiz_Score = c(78,85,65,92,70,88)  
)  
  
plot(data$Study_Time,  
  data$Quiz_Score,  
  cex=data$Videos_Watched/5,  
  pch=19,  
  col=rgb(0,0,1,0.5),  
  main="Study Time vs Quiz Score",  
  xlab="Study Time (Hours)",  
  ylab="Quiz Score")
```

Output:



Program 3-Monthly Average Quiz Score

```
data <- data.frame(  
  Student_ID = c("L01","L02","L03","L04","L05","L06"),  
  Quiz_Score = c(78,85,65,92,70,88),  
  Login_Date = c("2025-01-05","2025-01-05","2025-02-08",  
    "2025-02-08","2025-03-12","2025-03-12")  
)  
data$Login_Date <- as.Date(data$Login_Date)  
data$Month <- format(data$Login_Date,"%B")  
avg_score <- aggregate(Quiz_Score ~ Month, data, mean)  
print(avg_score)  
plot(avg_score$Quiz_Score,  
  type="o",  
  col="blue",  
  xaxt="n",  
  main="Average Monthly Quiz Score",  
  xlab="Month",  
  ylab="Average Quiz Score")  
  
axis(1, at=1:nrow(avg_score), labels=avg_score$Month)
```

Output:

